$f(x+y) = f(x) + f(y)$

$k\,f(x) = f(kx)$ $\quad f(x) = \frac{f(kx)}{k}$

$f\left(\frac{x}{k}\right) = \frac{f(x)}{k}$

$kf(x) = f(kx)$

$n! = n(n-1)!$

$f(x) = cx \quad \forall c \in \mathbb{Q}$

$f(x) = cx$

$V - E = 1$

$\frac{d}{dx} x^n = nx^{n-1}$

$x^n e^{-x}$ $\quad (x^n e^{-x})' = nx^{n-1}e^{-x}$

$V - E + F = 2$

$\underbrace{\quad}_{\to 0}$

$V - E + F = 2$

$\cancel{(x^n - 1)' = nx^{n-1}}$

$\int_0^\infty x^0 e^{-x}\,dx = \left[-e^{-x}\right]_0^\infty = 1$

$(x^n e^{-x})' = nx^{n-1}e^{-x} + x^n e^{-x}$ $\quad V - E + 1$

$\int_0^\infty x^n e^{-x}\,dx = n\int_0^\infty x^{n-1}e^{-x}\,dx$

$\cancel{x^n e^{-x} + (x^n e^{-x})' = nx^{n-1}e^{-x}}$

$\boxed{n! = \int_0^\infty x^n e^{-x}\,dx}$

$\sin x = x(x^2 - 1^2$

$\sin x = x\left(\frac{x^2}{\pi^2} - 1^2\right)\left(\frac{x^2}{\pi^2} - 2^2\right)\left(\frac{x^2}{\pi^2} - 3^2\right)\ldots$

$f(n) = 1 + nf(n-1)$

$(n \geq 1)$

$\int_a^b x^n e^{-x}\,dx + \underbrace{\left[x^n e^{-x}\right]_a^b}_{=0} = n\int_b^a x^{n-1}e^{-x}\,dx$

$a = 0, b = \infty$

HAI TIEN

67

$6+7=13$

$1+3=4$

99999999999

↓

9·9

↓

18

↓

9

99

↓

18

↓

9

999

↓

27

↓

9

67

1 ← 10 ← 19 ← 199 ← 1999...

x ?

10 ≠ P

~~.: 14262163p~~

M

$f(x) = x f(x-1)$

$\cancel{f'(x) = 1 + x f'}$

$\cancel{f'(x) = f(x-1) +}$

$6 \{ \ldots a_n \ldots a_2 a_1 a_0 \}_p$

$\downarrow$

$\cancel{a_n + p a_1}$

$a_0 + p a_1 + p^2 a_2 + \ldots + p^n a_n + \ldots$

$f'(x) = f(x-1) + x f'(x-1)$

$f''(x) = 2 f'(x-1) + x f''(x-1)$

67

PP

1

ϕ

ϕ^2

$\sqrt{2}$

$\sqrt{2}\phi$

$\sqrt{2}\phi^2$

$\sqrt{3}$

$\sqrt{3}\phi$

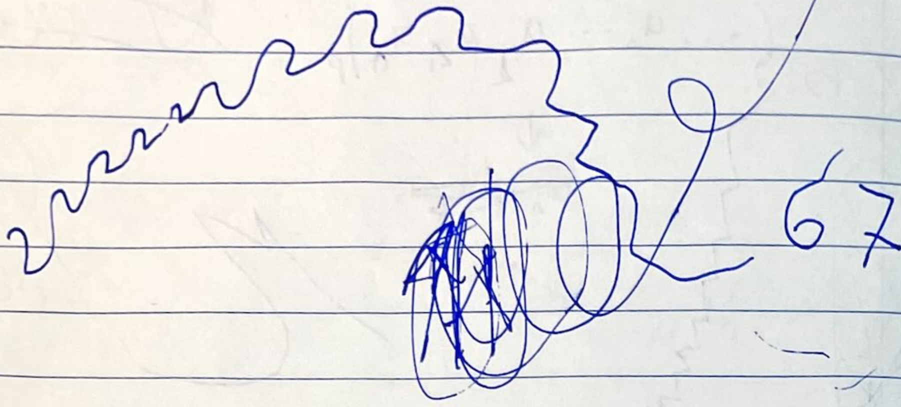
$\sqrt{3}\phi^2$

$\sqrt{5+\sqrt{5}}$

$\sqrt{5+\sqrt{5}}\phi$

$\sqrt{5+\sqrt{5}}\phi^2$

ϕ ψ



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$f(0) = 1$

$f(1) = f(0)$

$f(1) = f(0) + f'(0)$

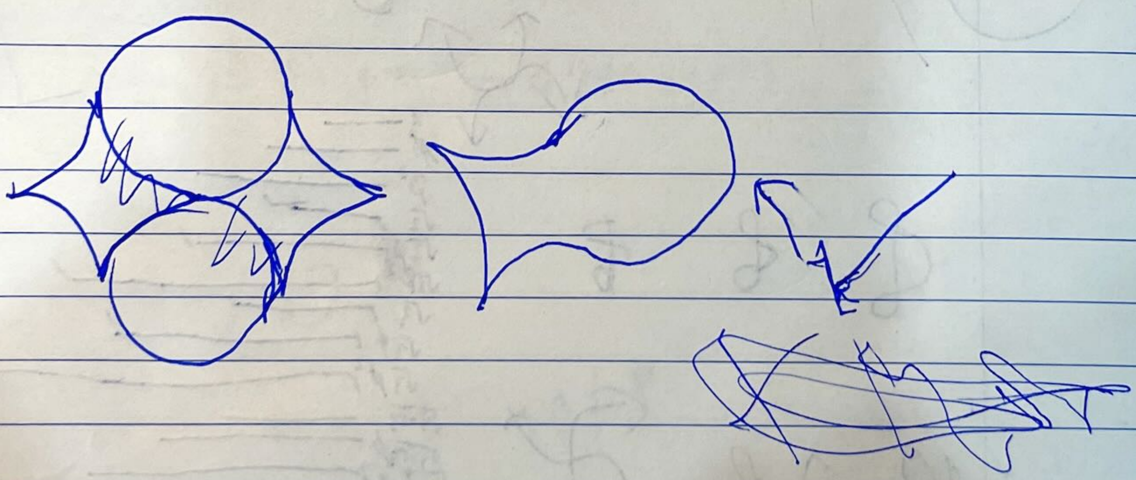
$f'(1) = f(0) + f''(0)$

6 $\frac{9}{1}$

$f''(1) = 2f'(0) + \dots$

$f^{(n-1)}(1) = (n-1) f^{(n-2)}(0) + f^{(n-1)}(0)$

$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$



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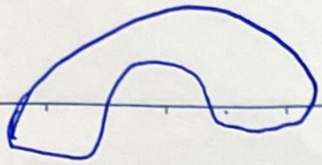

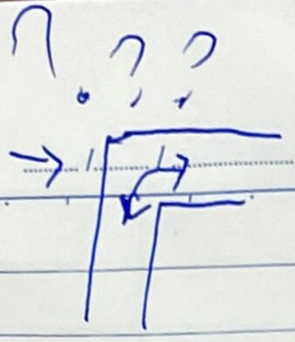

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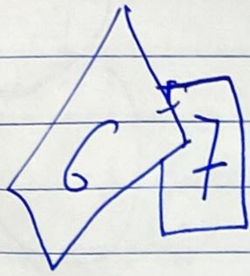
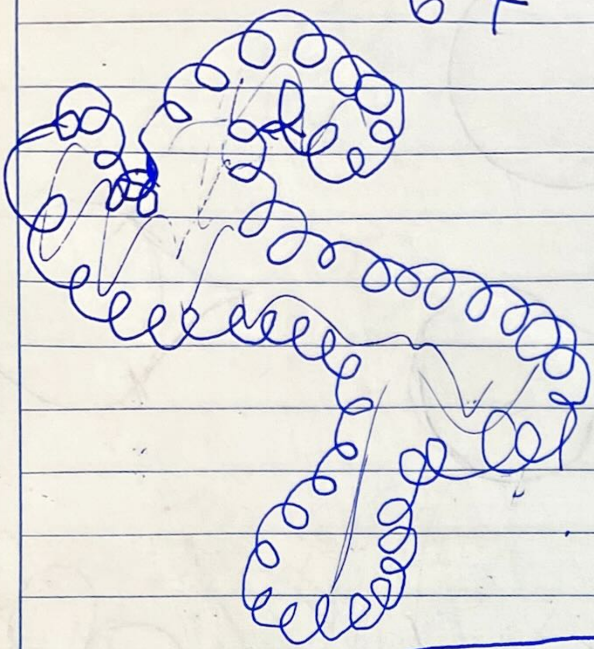
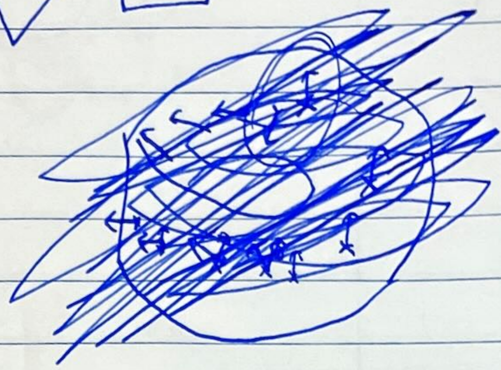
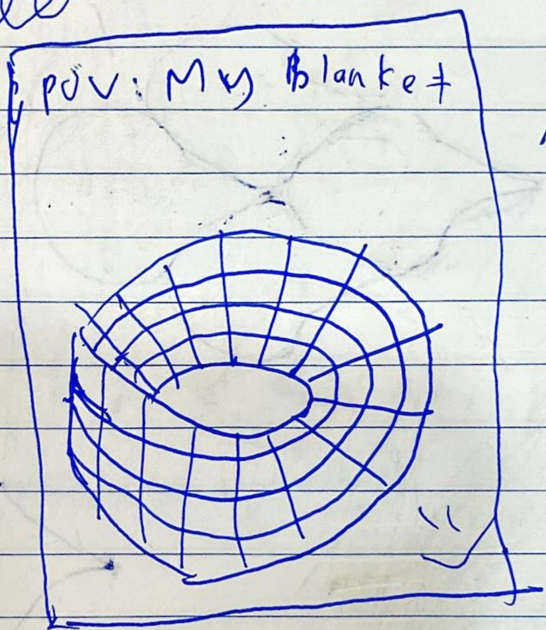
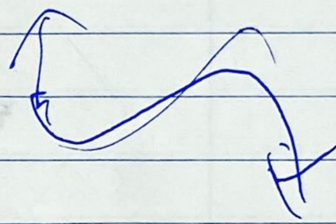
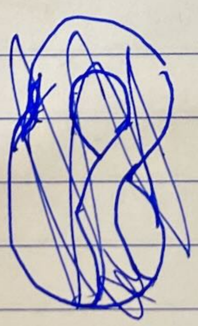


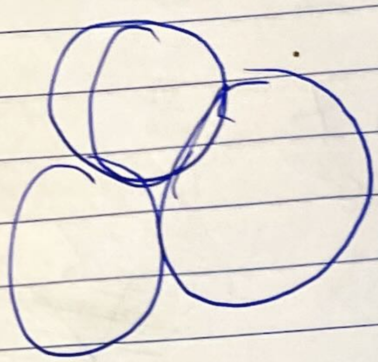


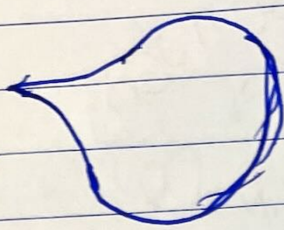


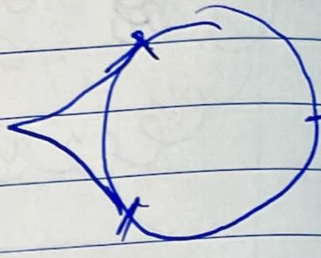


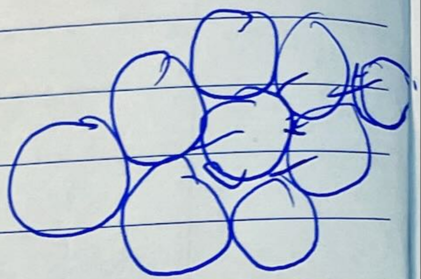


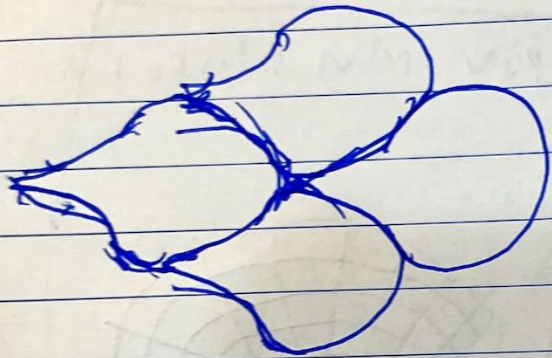


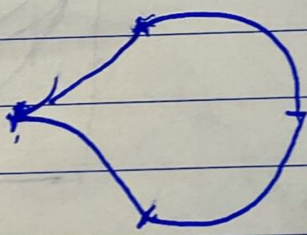


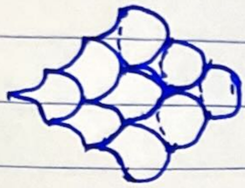


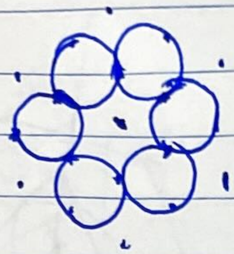


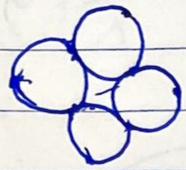


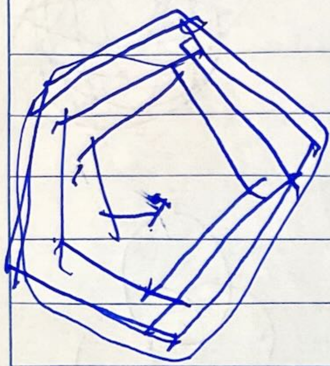


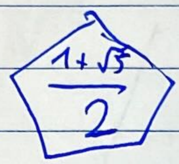


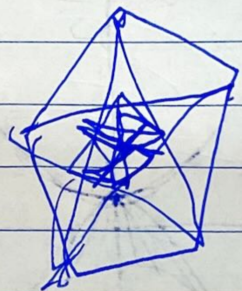


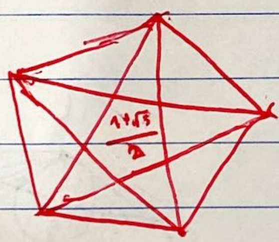




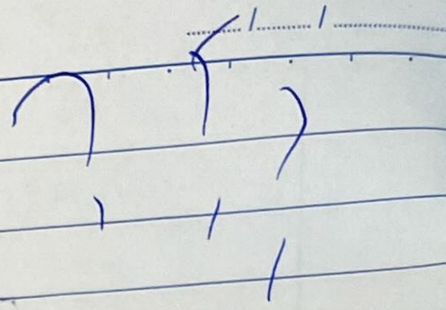



$\frac{1+\sqrt{5}}{2}$

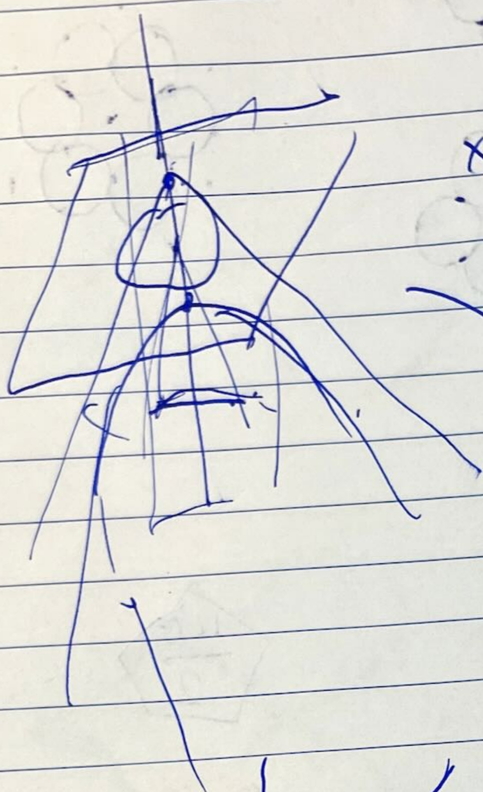


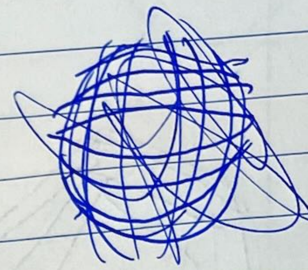

$\frac{1+\sqrt{5}}{2}$

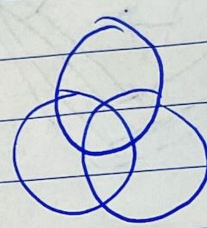
$$\begin{pmatrix} \cosh\theta & \sinh\theta & 0 \\ \sinh\theta & \cosh\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



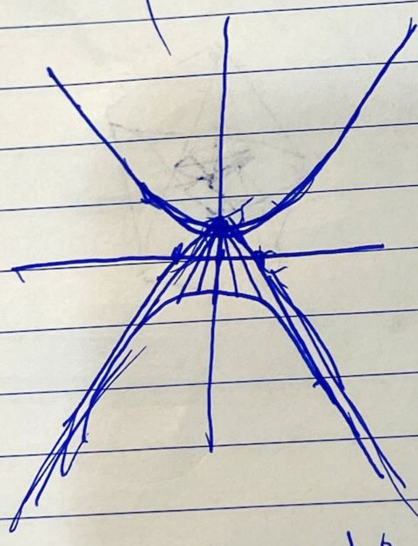
$x^2+y^2-z^2=1$



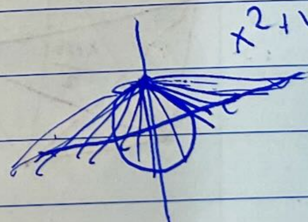




$x^2-y^2=1$



$x^2+y^2=1$



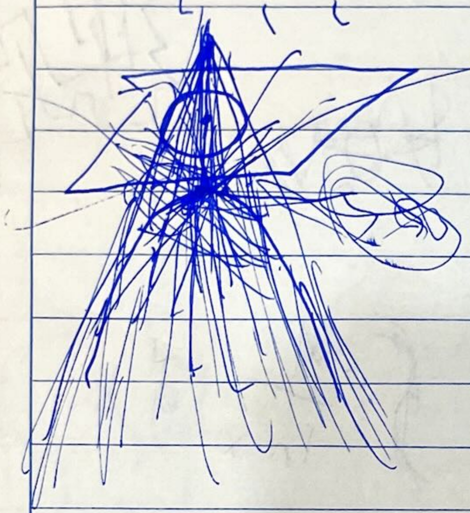
$$\begin{pmatrix} \cosh\theta & \sinh\theta \\ \sinh\theta & \cosh\theta \end{pmatrix} \xleftarrow{?} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

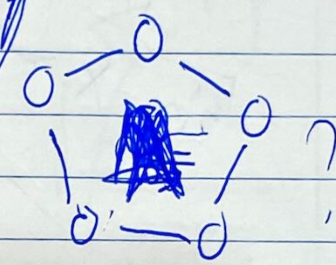
$$\sinh\theta = \frac{e^{\theta} - e^{-\theta}}{2} \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$\cosh\theta = \frac{e^{\theta} + e^{-\theta}}{2} \qquad \cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\sinh\theta = \frac{e^{j\theta} - e^{-j\theta}}{2} \qquad j^2 = 1,\ j \neq \pm 1$$

$$\cosh\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$





NASINAZ

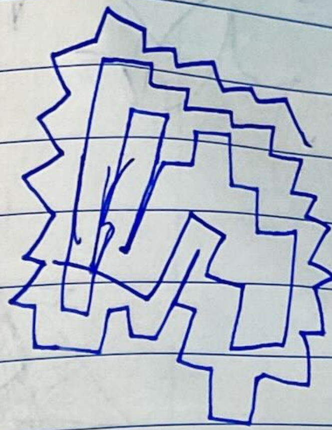
$\sin x = 2i$

$\frac{e^{ix} - e^{-ix}}{2i} = 2i$

$e^{ix} = y$

$y - \frac{1}{y} = 2ia$

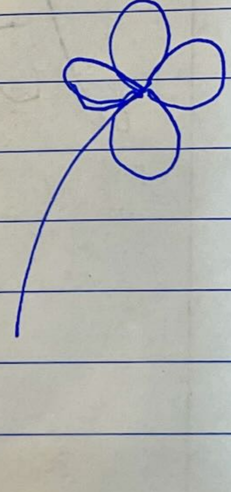
$y^2 - 2iay - 1 = 0$



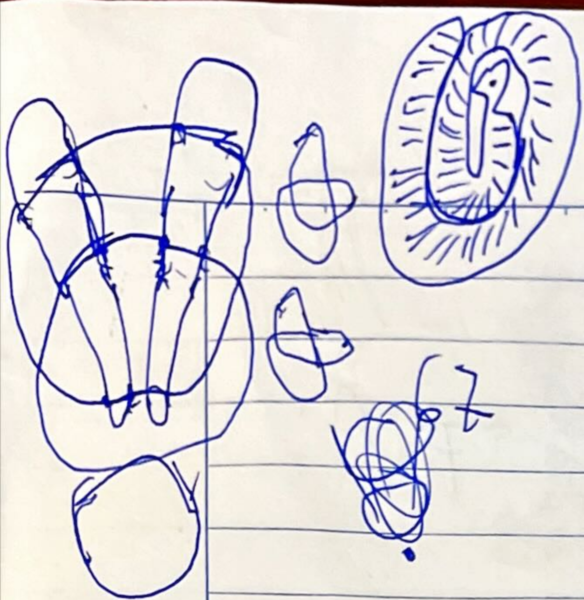
$\sin x = 2$

$\int \frac{1}{\ln x} \, dx$

$f(x)\Big|_0^x$



$f(x)$



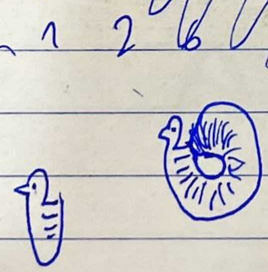
dy

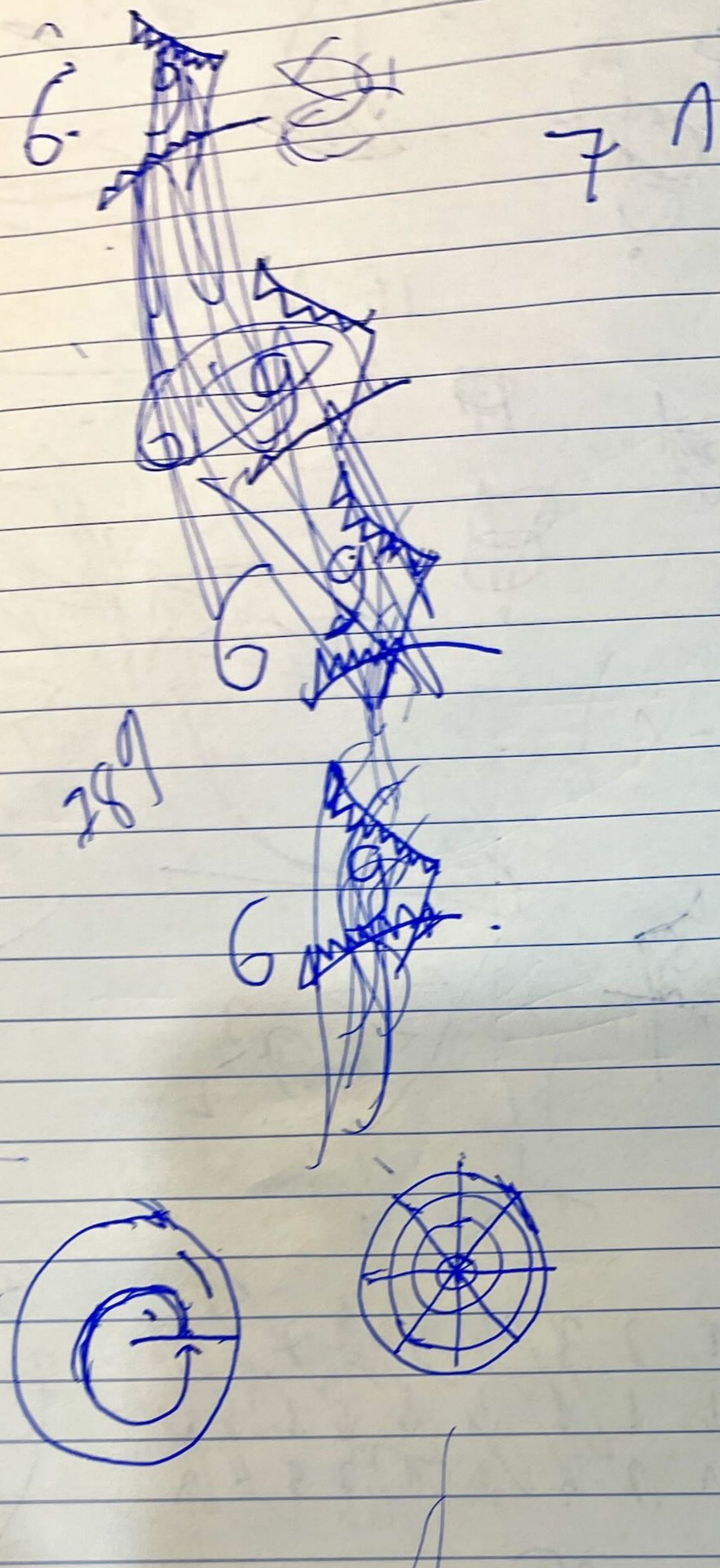
IFON

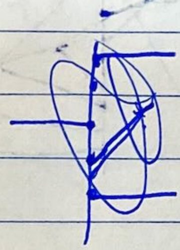
$\int \left(x^{dx} - 1 \right)$

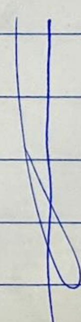
$2^{2}{}_{2}$

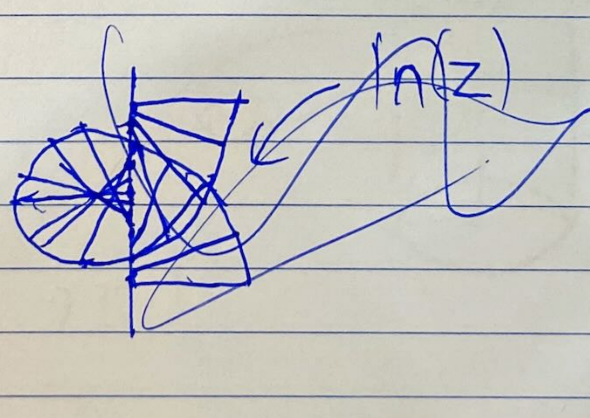
0	1	2	3	4	5	6	7	8	9
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
2	1	2	6	2	1	7	5	4	3



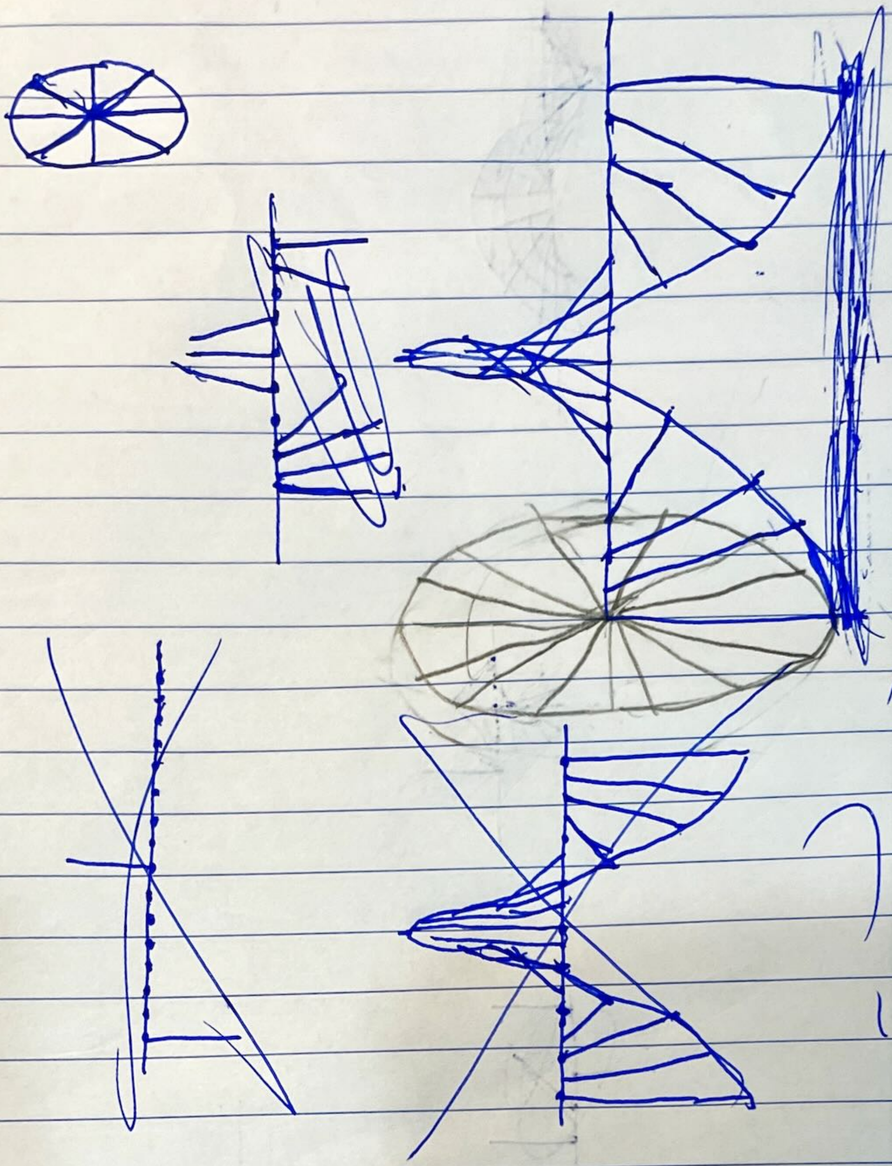

6.
7 1
6
789
6

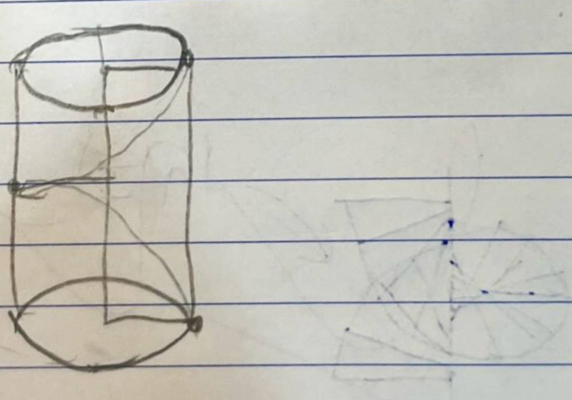





ln(z)

In 2

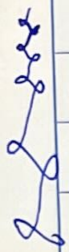


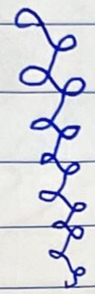


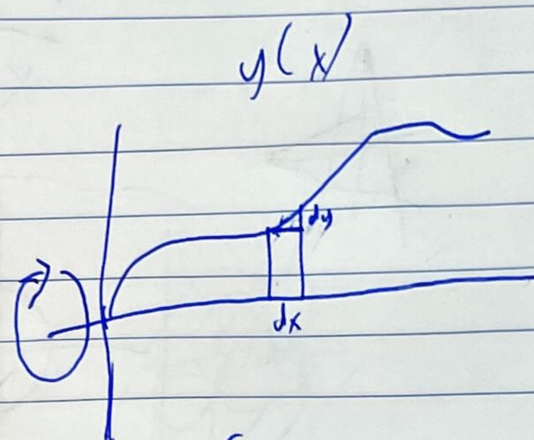
1 2 2 4 2 4 4 8 2 4 4 8 4 8 8 16

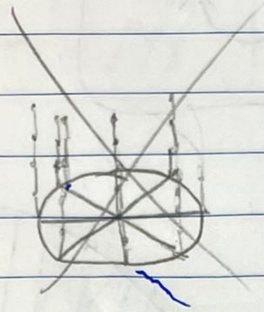
$re^{i\theta}$ mod 7

1 2 2 4 2 4 4 1 2 4 4 1 4 1 1 2

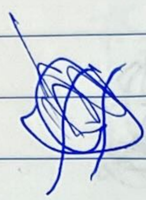








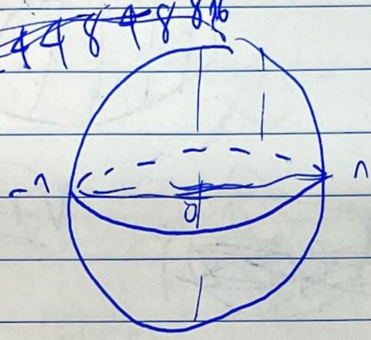
$$\int 2\pi y \sqrt{dx^2+dy^2}$$

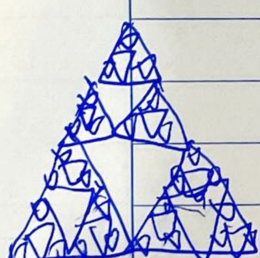


$$\int 2\pi y \sqrt{1+\left(\frac{dy}{dx}\right)^2}\, dx$$

$$2\pi \int y \sqrt{1+\left(\frac{dy}{dx}\right)^2}\, dx$$

1 2 2 4 2 4 4 8 2 4 4 8 4 8 8 16



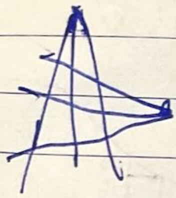


$$2\pi \int^{1} \sqrt{1-x^2}\,\sqrt{1+\frac{-2x}{\sqrt{1-x^2}}}\, dx$$

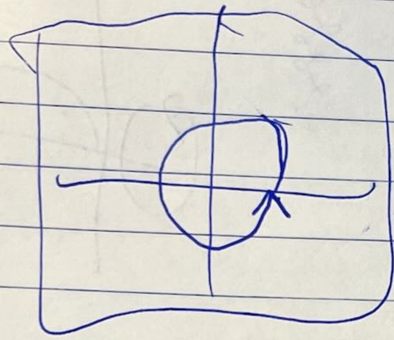
4π

$$2\pi \int_{-1}^{1}$$

......./......./.........



$\frac{\wedge}{X}$



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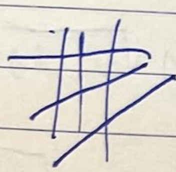
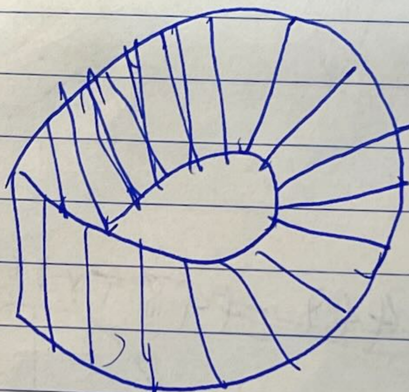




$\cancel{\forall(r;\theta); r=1 \Rightarrow \theta}$

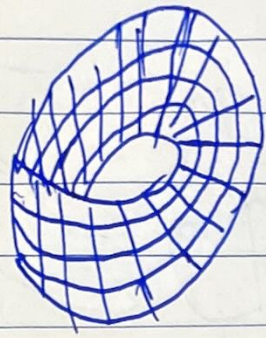
$r;\theta$

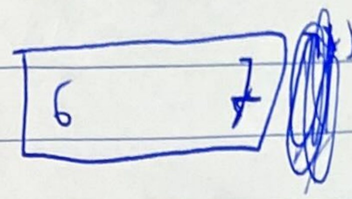
$\forall(r;\theta); r=1 \Rightarrow \text{line } \frac{\theta}{2}$

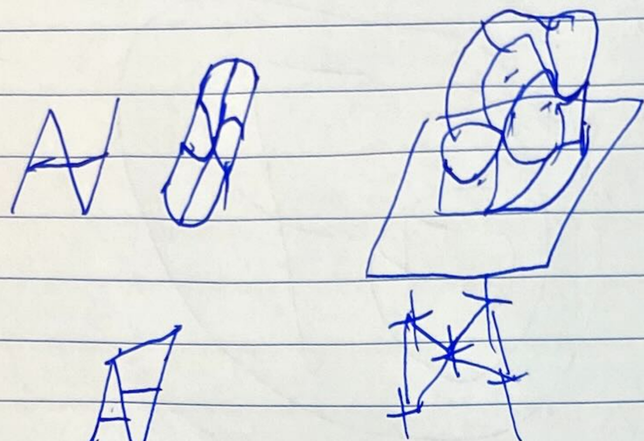
0 5 10 15 20

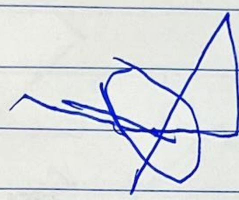
HAI TIEN

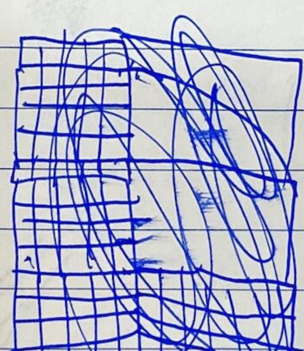
?

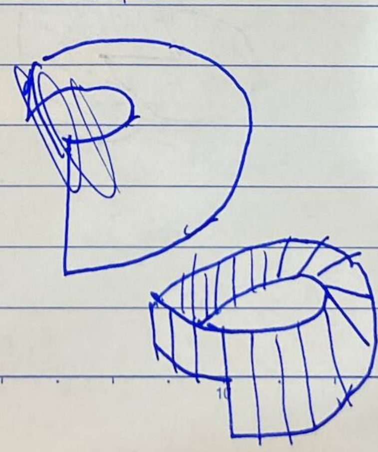



6
7

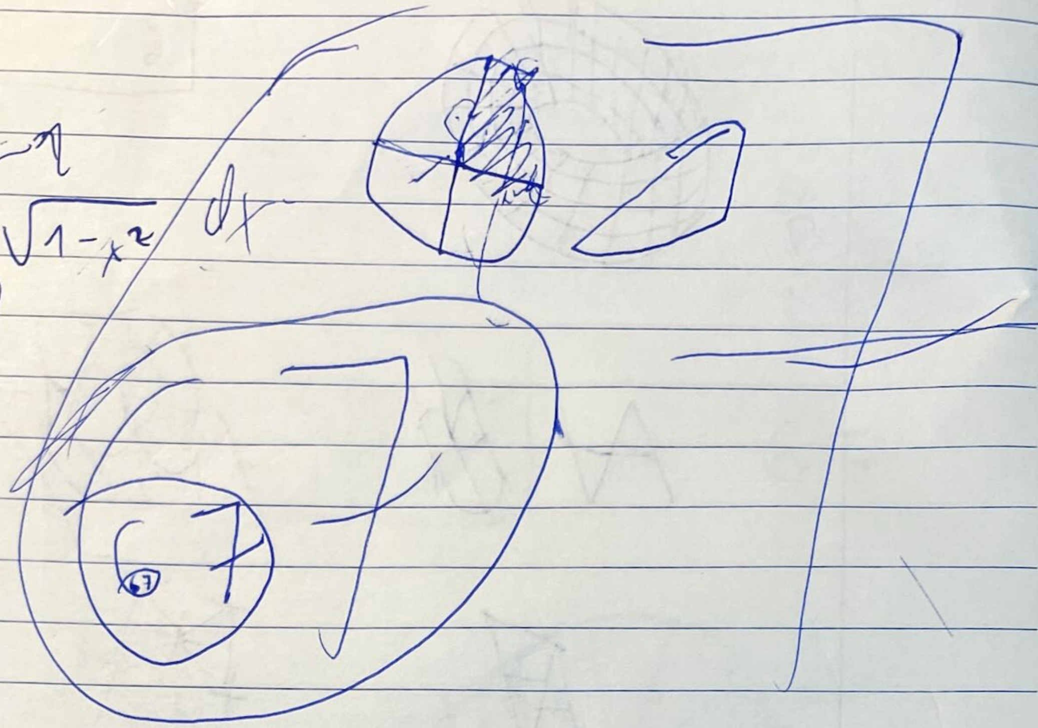


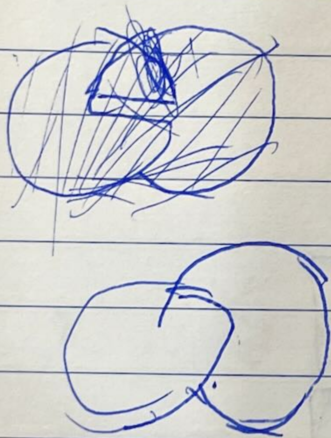


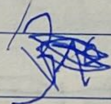


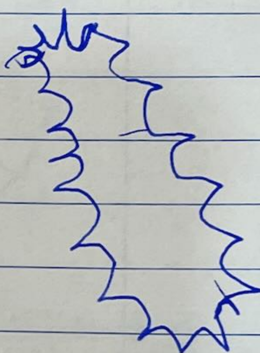


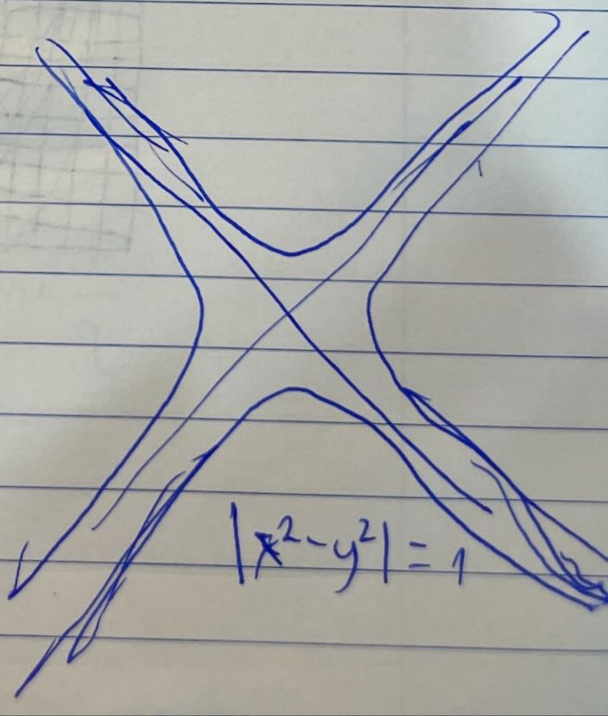
$$\int_0^1 \sqrt{1-x^2}\,dx$$











$$|x^2-y^2|=1$$

67

67

$f'(x) = f(x + \frac{\pi}{2})$

67

67

3.14159265

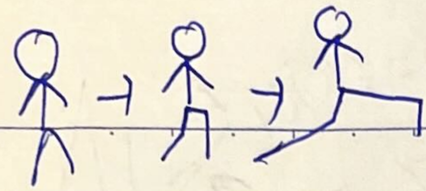
6

X

3

0 5 10 15 20

HAI TIEN



$f'(x) = f(x$

$''(x) = f(x + \frac{\pi}{2})$

$67^{67^{67}}$

$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)\, x^n}{n!}$

$= \sum_{n=0}^{\infty} \frac{f(\frac{\pi}{2})\, x^n}{n!}$

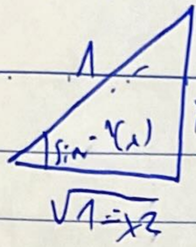
Japan

made in China

$$\boxed{\int \sqrt{1-x^2}\, dx = \frac{\sin^{-1}(x) + x\sqrt{1-x^2}}{2} + C}$$

HAI TIEN

$$\int (1-\sin^2 x)\,dx = \sin x \cos x + \int \sin^2 x \, dx$$



$$2\int \sin^2 x \, dx = x - \sin x \cos x$$

$$\int_0^{\frac{\pi}{2}} (1-\sin^2 u)\,du$$

$6+7=13$

$$\int_0^{\frac{\pi}{2}} \cos^2 u \, du$$

$$\left[\frac{x - \sin x \cos x}{2}\right]_0^{\frac{\pi}{2}}$$

$A = \pi r^2$

$\frac{\pi}{2} : \frac{\pi}{4}$

$$\int_0^1 \sqrt{1-x^2}\,dx \quad 67 \quad \int_0^{\frac{\pi}{2}} \cos^2 x \, dx$$

6767

$$\boxed{\int_0^1 \sqrt{1-x^2}\,dx = \frac{\pi}{4}}$$

$\sin^{-1}(x)$

$\boxed{u = \sin^{-1}(x)}$

$u = \sin^{-1}(x)$

$\cos x \mid \frac{\sin x}{\cos x}$

$x = \sin(u) - \sin x \mid \cos x$

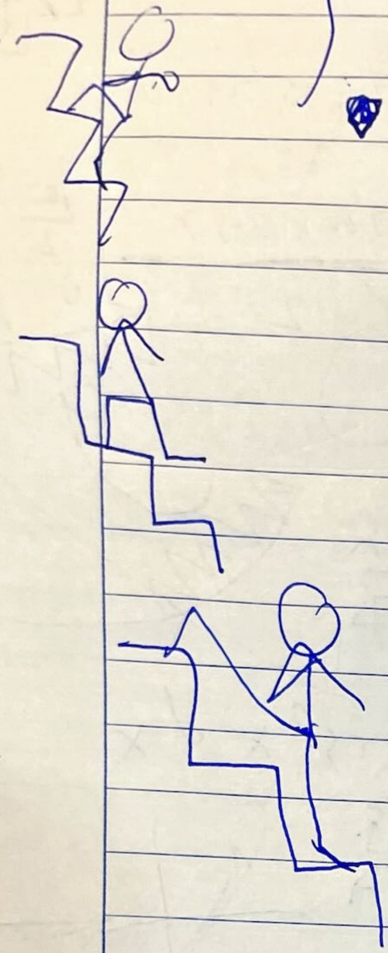
$\cancel{A = \pi r^2}$

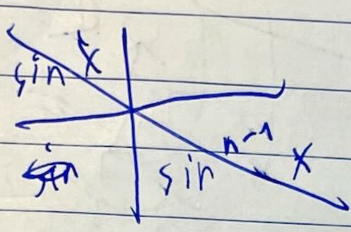
$dx = \cos(u)\,du$

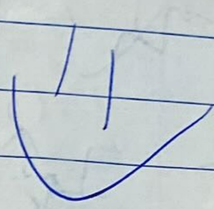
$$dx = \cos(\sin^{-1}(x))\,du = \sqrt{1-x^2}\,du$$

$$\int_0^{\frac{\pi}{2}} (1-x^2)\,du \int \cancel{(1-\sin^2 x)\,dx}\ \sin x \cos x + \int \sin^2 x \, dx$$

$$\int \sin^n(x)\,dx$$







$-\cos x$	$\sin^{n-1} x$
$\sin x$	$(n-1)\sin^{n-2}x\cos x$

$$-\cos x \sin^{n-1}x-(n-1)\int \sin^{n-2}x\cos^2 x\,dx$$

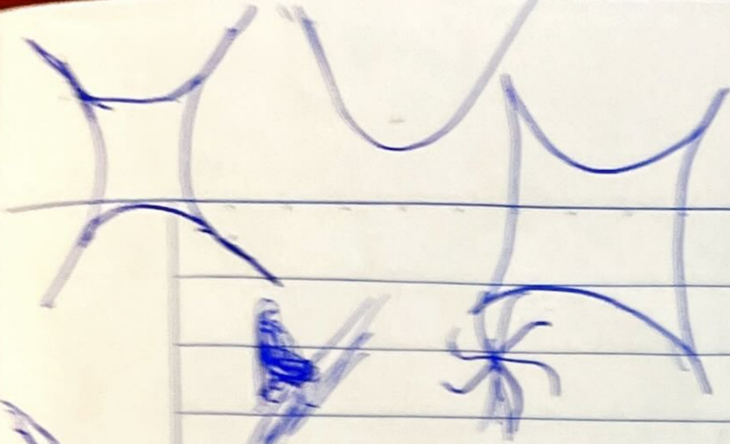
$$-\cos x\sin^{n-1}x-(n-1)\int(\sin^{n-2}x-\sin^n x)\,dx$$

$$A_n=-\cos x\sin^{n-1}x-(n-1)(A_{n-2}-A_n)$$

$$(2-n)A_n=-\cos x\sin^{n-1}x-(n-1)A_{n-2}$$

$$(n-2)A_n=\cos x\sin^{n-1}x+(n-1)A_{n-2}$$

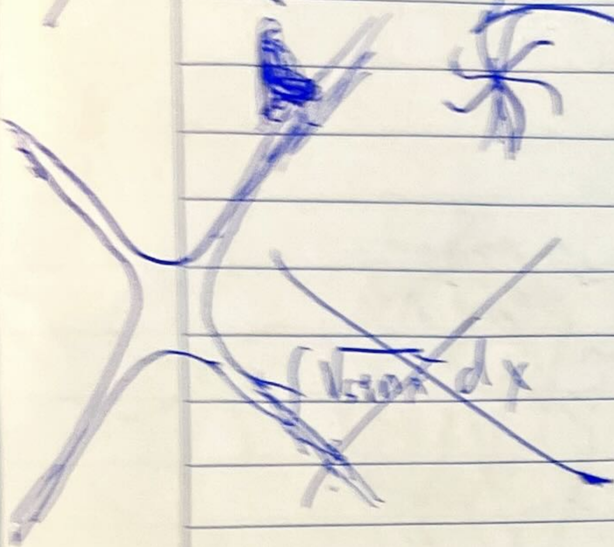
$$A_n=\frac{\cos x\sin^{n-1}x}{n-2}+\frac{n-1}{n-2}A_{n-2}$$



$i^2 = -1 \to$

$\varepsilon^2 = 0 \to$

$j^2 = 1 \to$

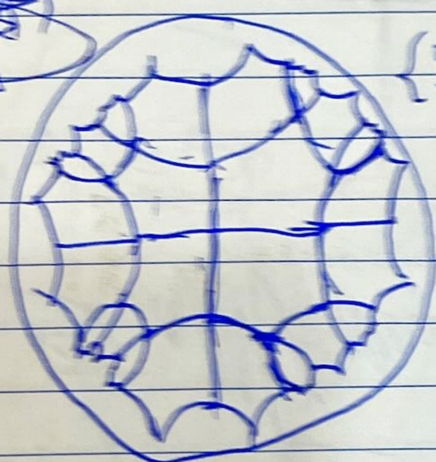


$\cancel{\int \sqrt{\sin x}\, dx}$

$\ln(\varphi + \sqrt{\varphi})$

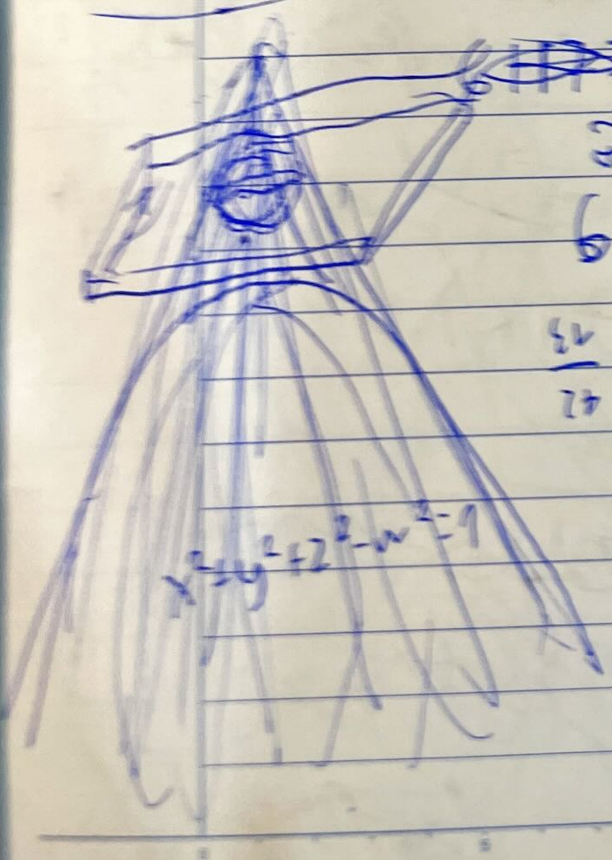
$\{5;4\}$

$\{5;3;4\}$



skibidi

$\cancel{6117}$



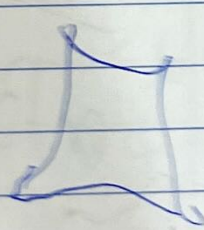
$x^2+y^2+z^2-w^2=1$

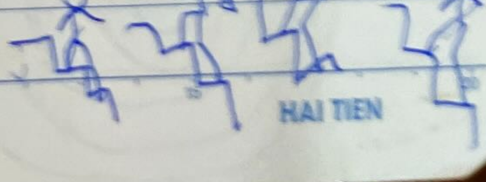
$\cancel{6117}$

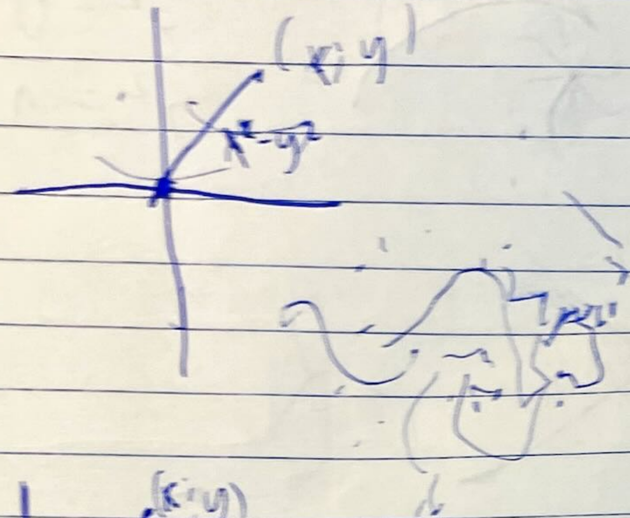
$6 \parallel 7 = \frac{42}{13}$

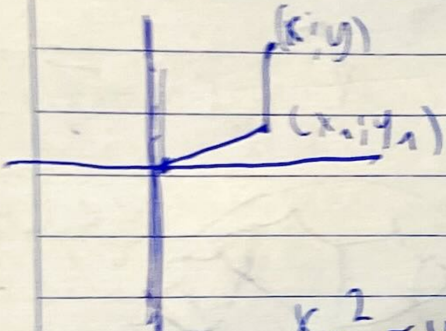
$\frac{7}{7}$

$\frac{4}{3}$

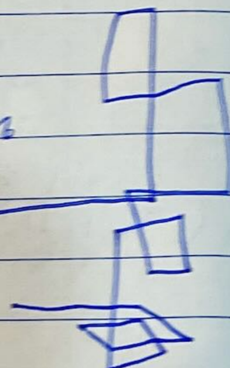








$x_1^2 - y_1^2 + (x-x_1)^2 + (y-y_1)^3$

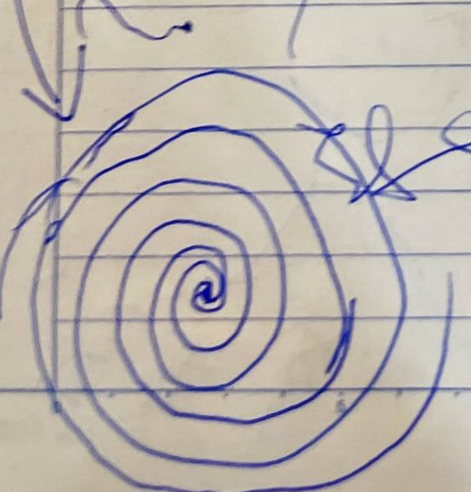


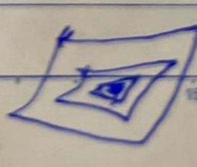
$\cancel{x^2} - 2xx_1 + 2x_1^2 - \cancel{y^2} + 2yy_1 + 2y_1^2$

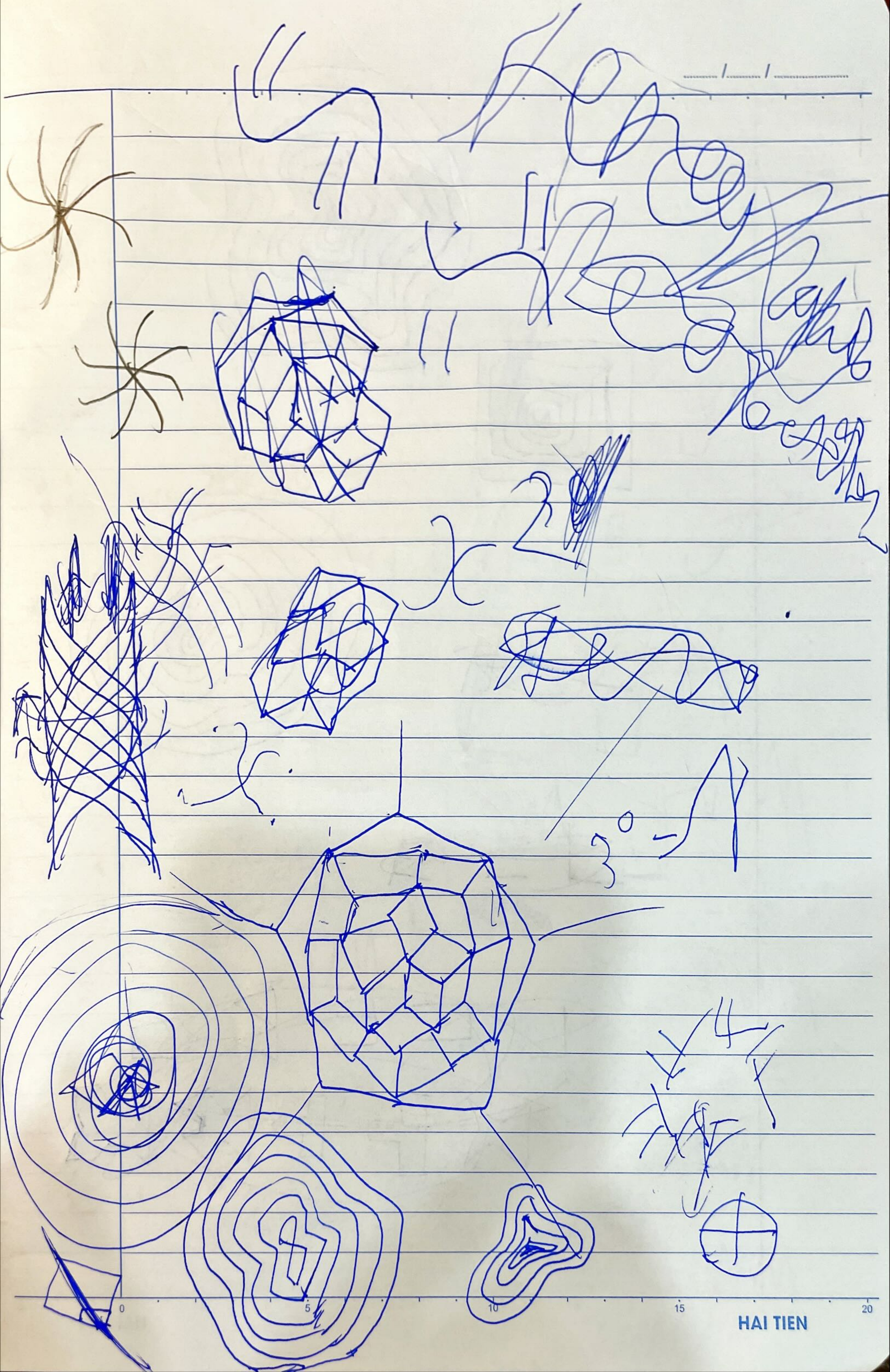
1 2 3 4 5 6 7 8 9 10

2x spiral

α β γ δ ε ζ η θ ι κ λ μ







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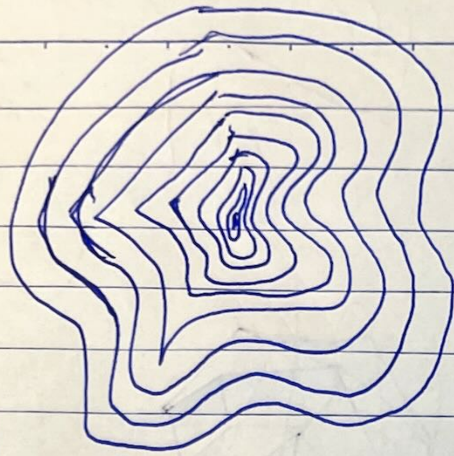
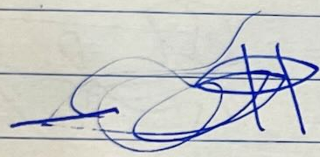
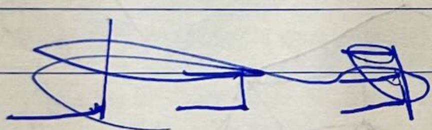
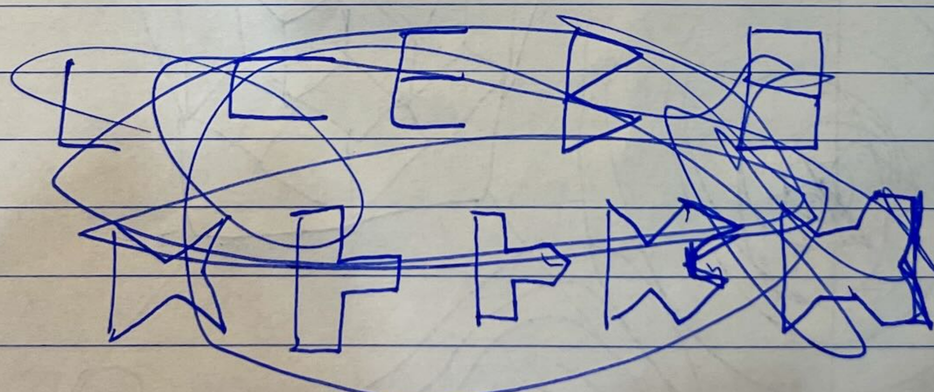
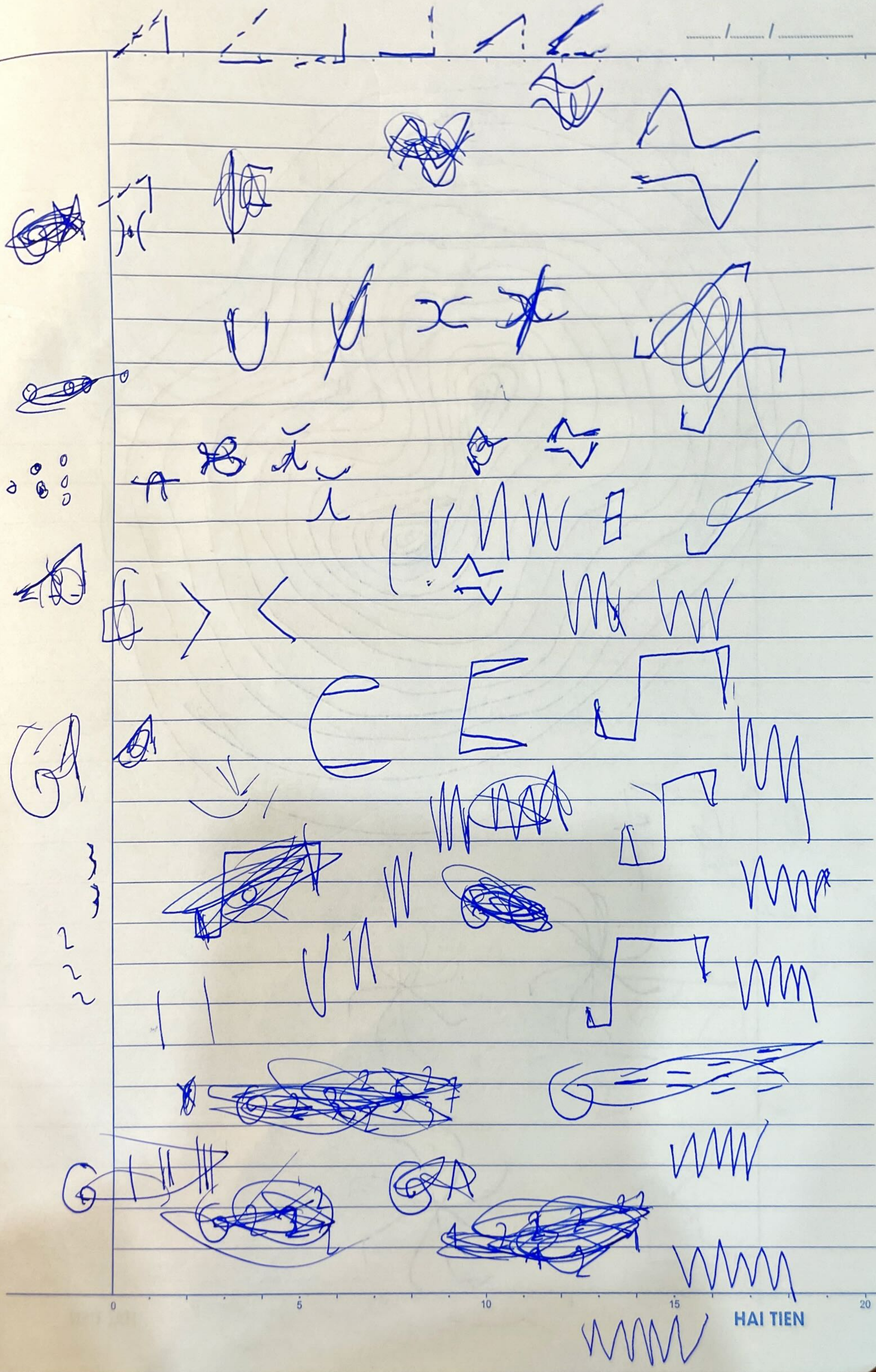
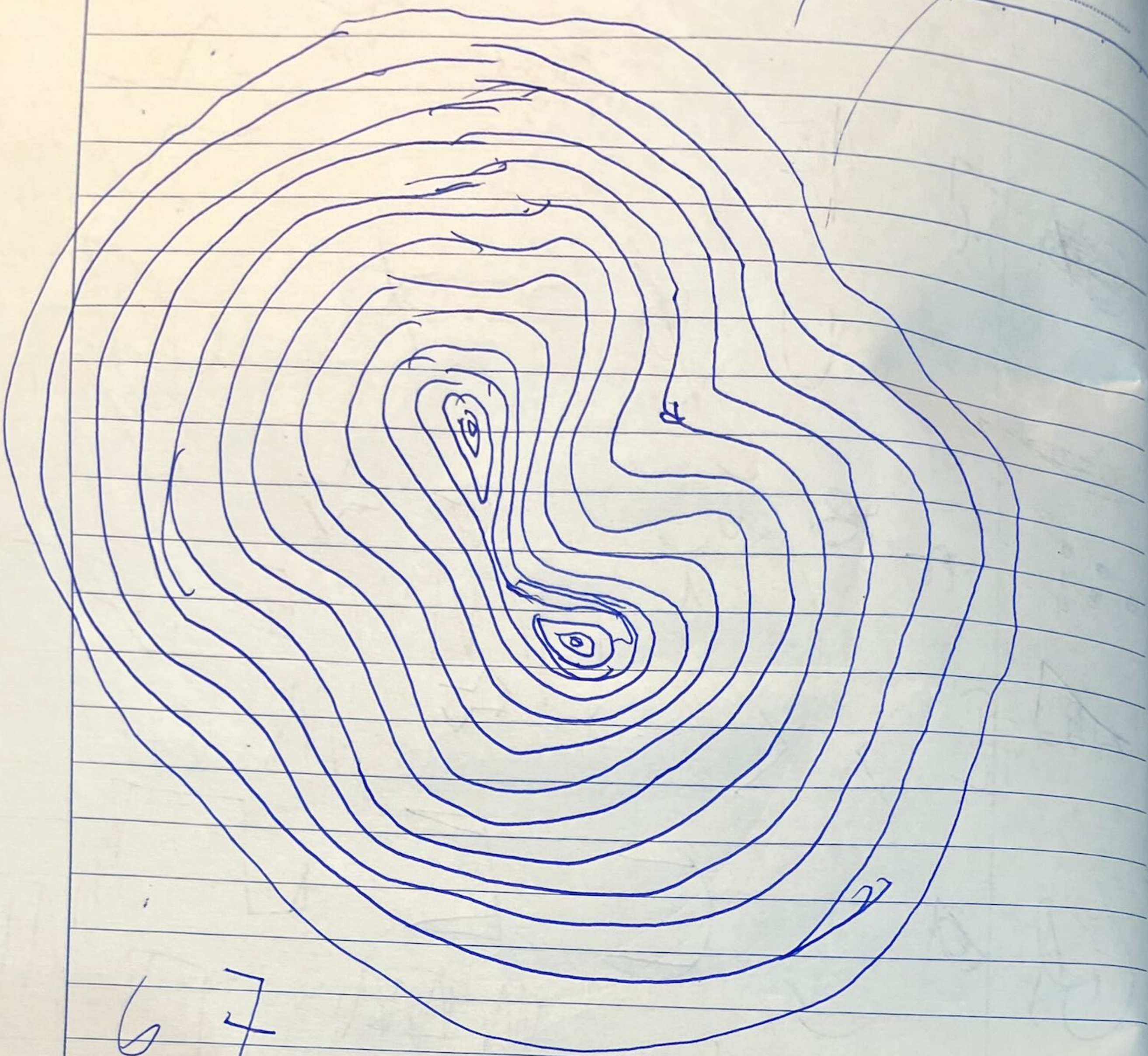




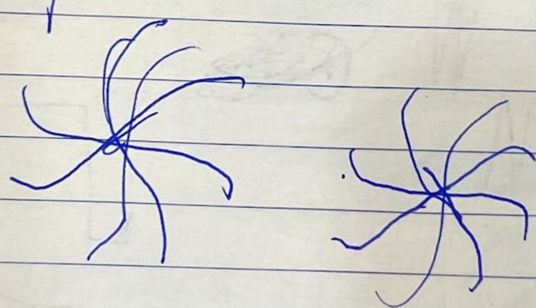


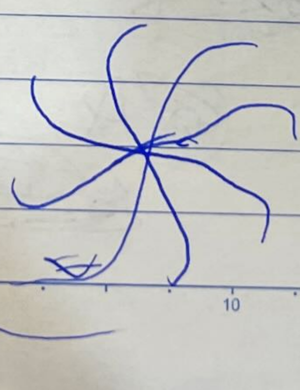


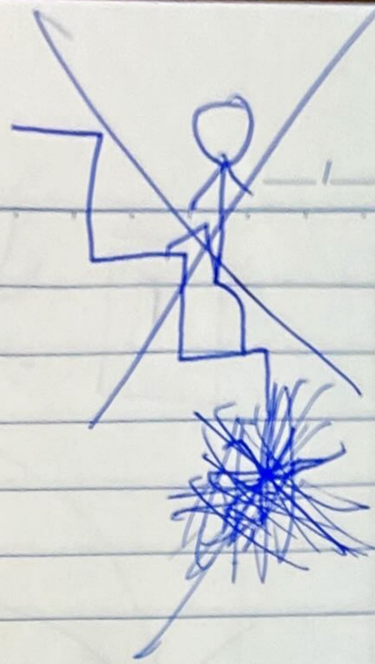




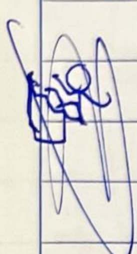
6 7

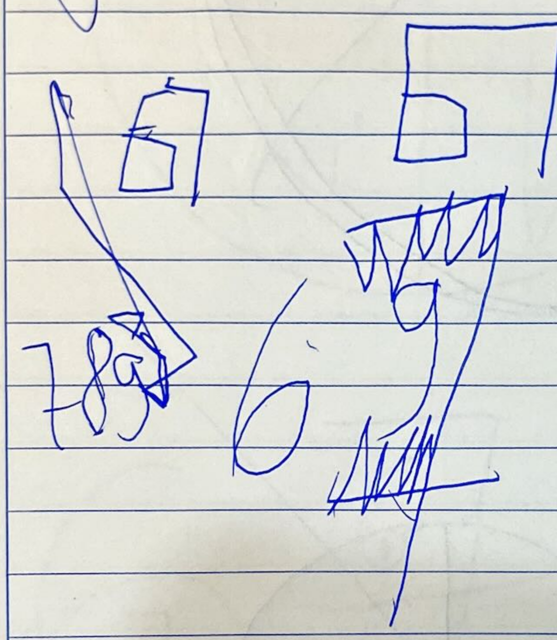


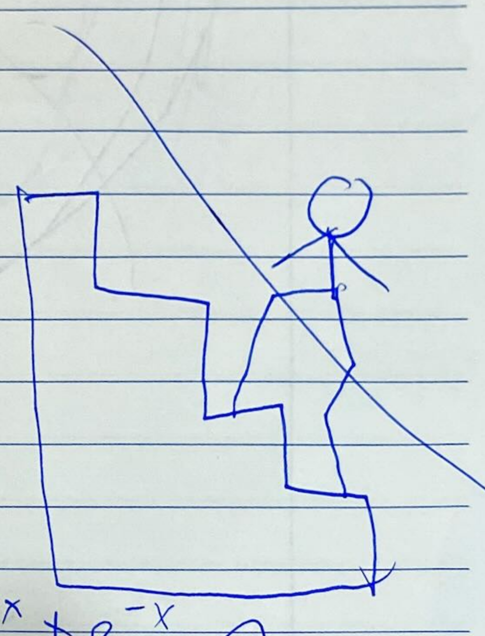




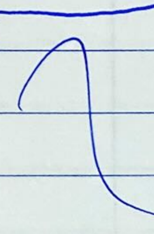
$$\int \sqrt{x^3+1}\, dx$$



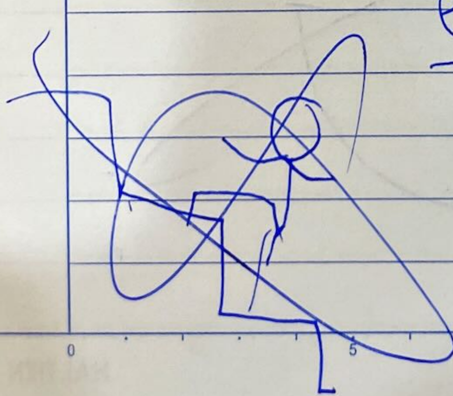


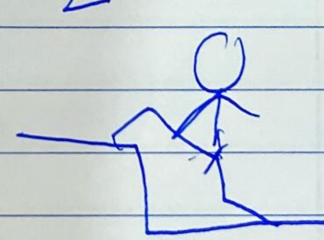


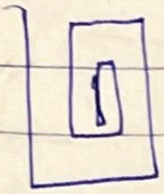
$$\frac{e^x + e^{-x}}{2}$$

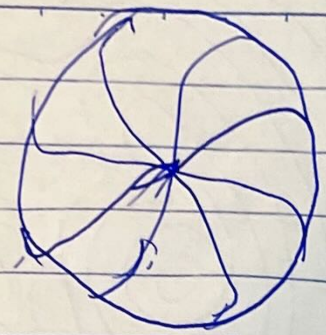


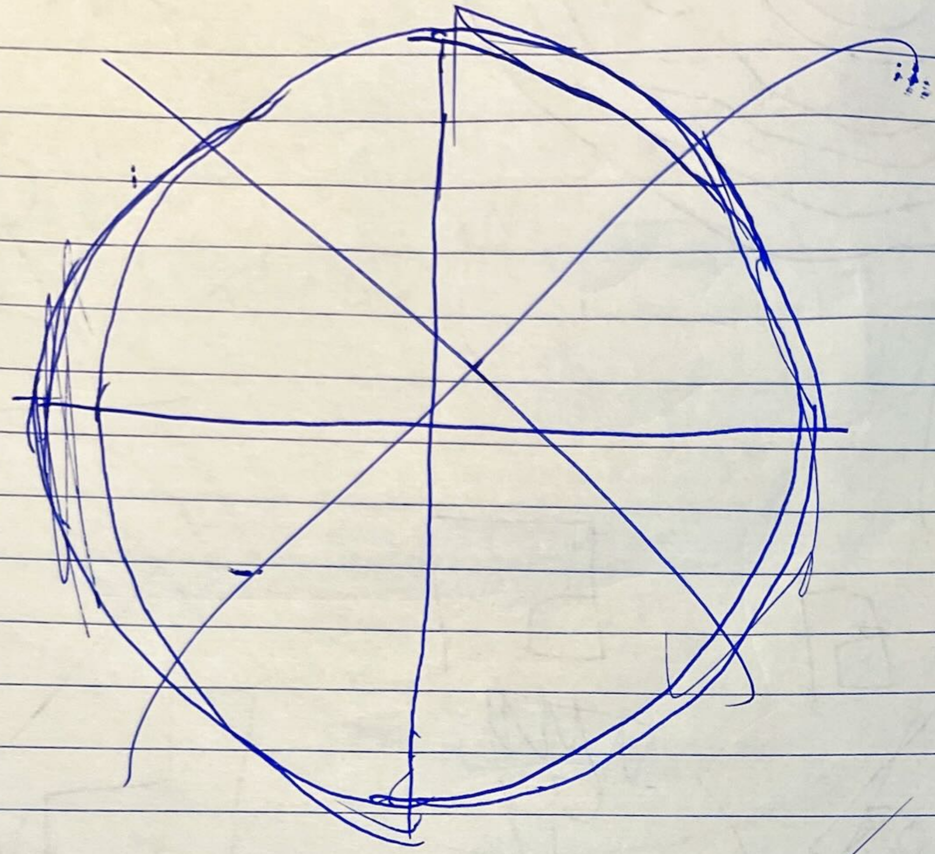
$$\frac{e^x - e^{-x}}{2}$$

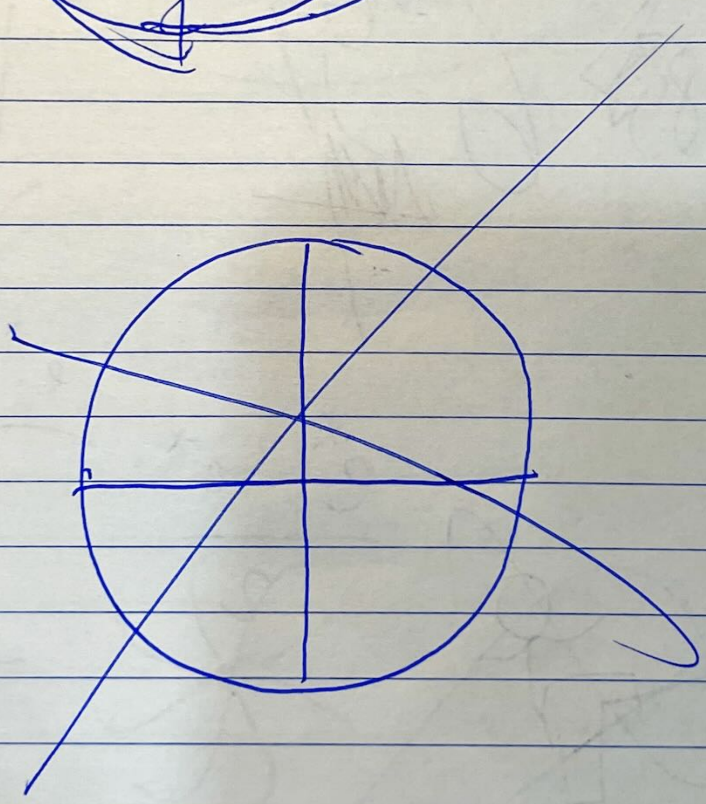


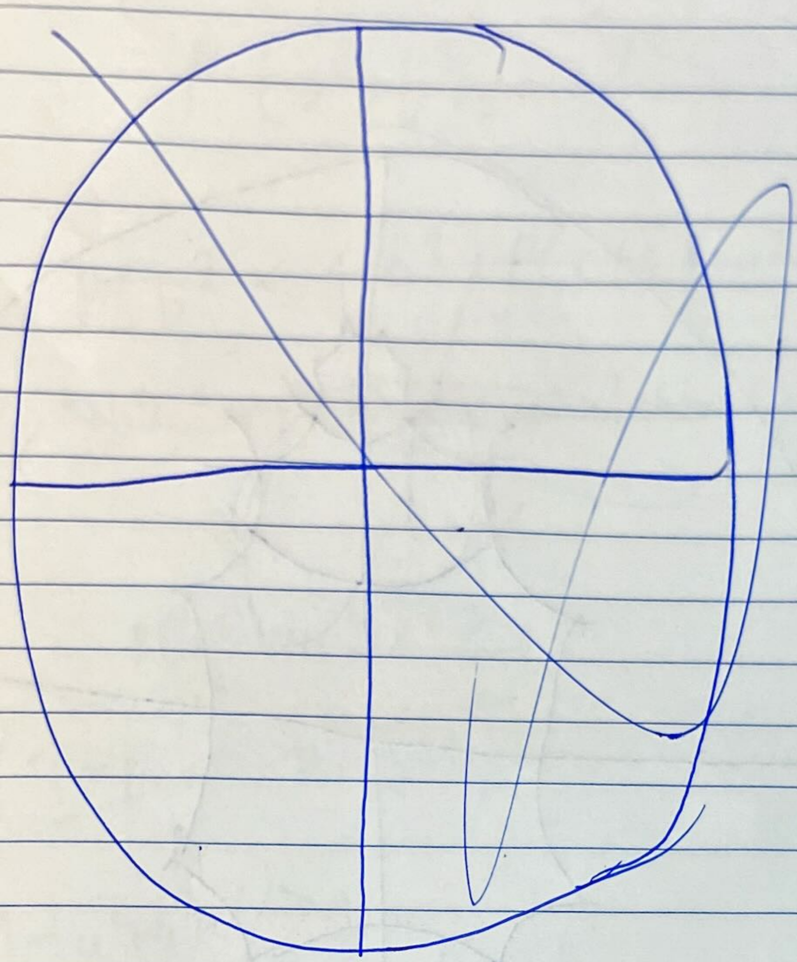


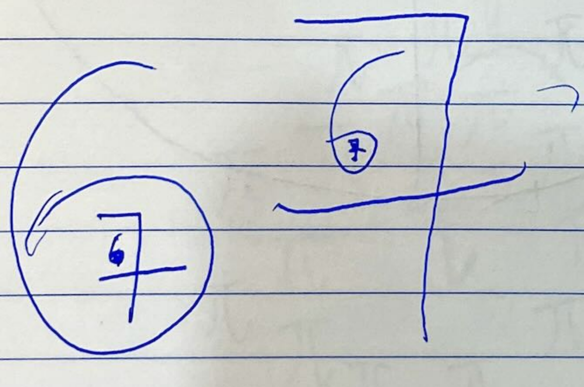


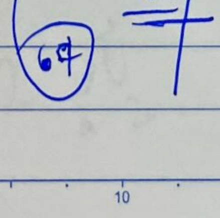


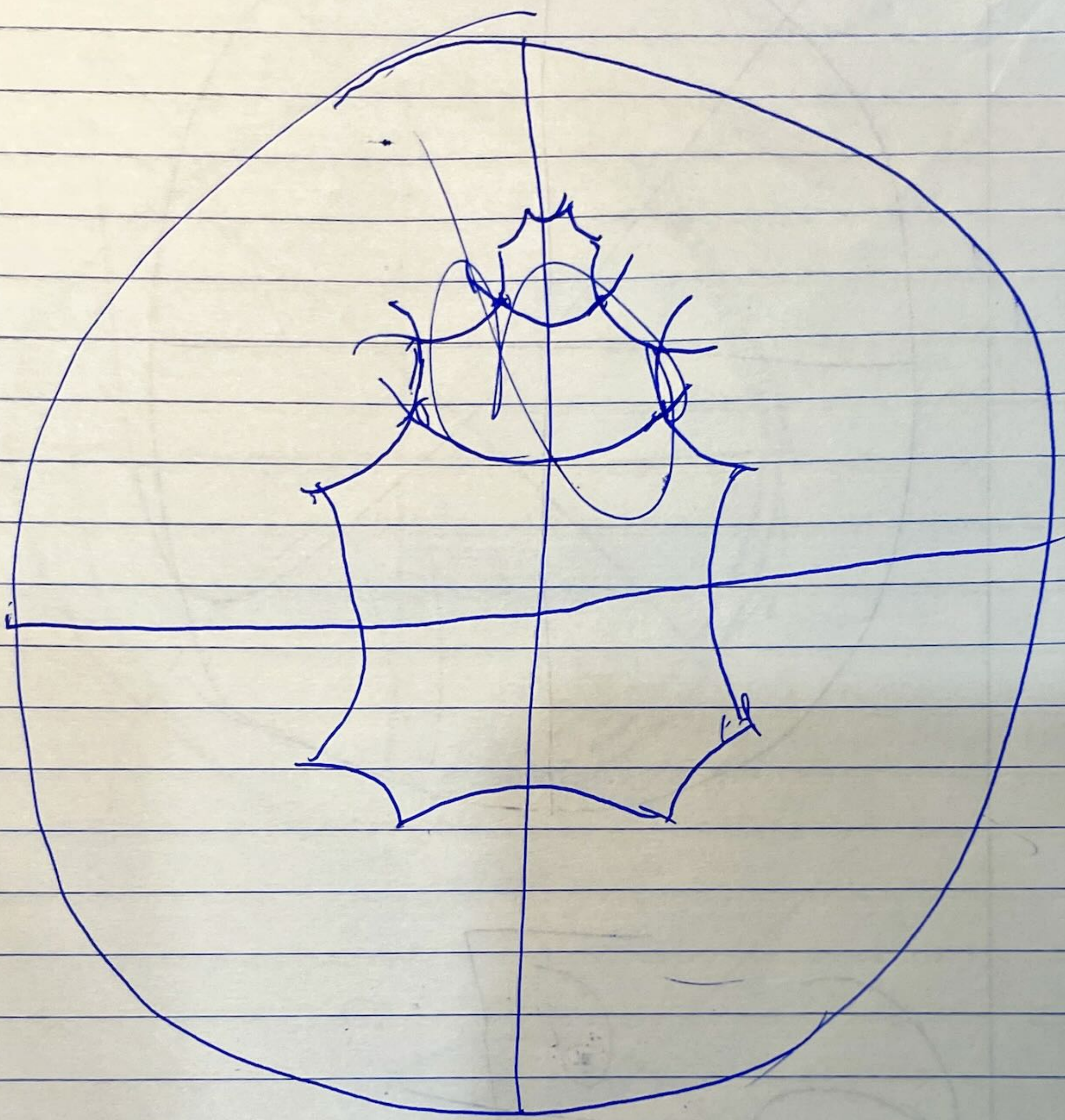












$a; b; c; d = \text{rand}(0; 1)$

$P(a^2+b^2+c^2+d^2<1) = ?$

$\cancel{P(b^2 < 1-a^2)\,P(c^2 < 1-a^2-b^2)\,P(d^2 < 1-a^2-b^2)}$

$\cancel{P(a^2+b^2+c^2<1)\,P(d^2<1-a^2-b^2-c^2)}$

$\cancel{P(a^2+b^2+c^2+d^2 \geq 1)}$

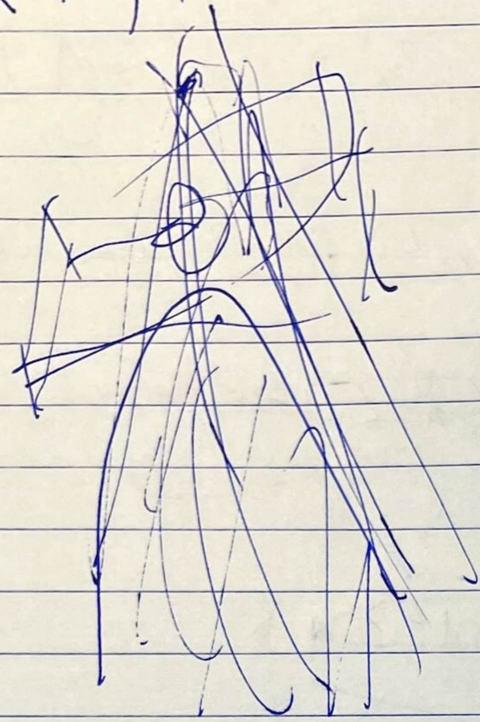
$\cancel{(1-P(a^2+b^2+c^2\geq 1))}$

$1-(a^2+b^2+c^2)>0$

π π π
π π π π
π π π π π
π π π π π
π π π π π
π π π
π π π

$\tau_1 = 2\pi^2$

$$x^2+y^2+z^2-w^2=1$$



$$\frac{1}{r} \overset{?}{\longrightarrow} \frac{1}{\sinh r}$$

sin cos

$$\frac{x}{w};\ \frac{y}{w};\ \frac{z}{w}$$

$$\Downarrow ?$$

$$\frac{x}{\sinh w};\ \frac{y}{\sinh w};\ \frac{z}{\sinh w}$$

sin
cos
sinr
cosr
sinh
cosh

$\operatorname{sinr}(x) = x$

$\operatorname{cosr}(x) = 1$

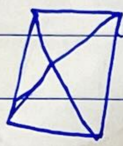
$\varepsilon^2 = 0$

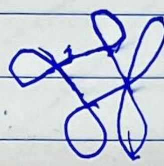
$\varepsilon \neq 0$

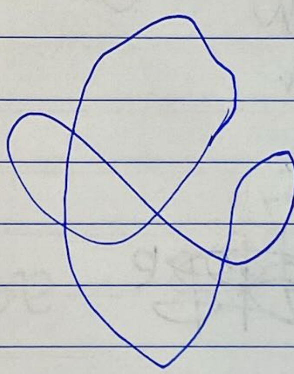
$i \; ; \; \varepsilon \; ; \; j$

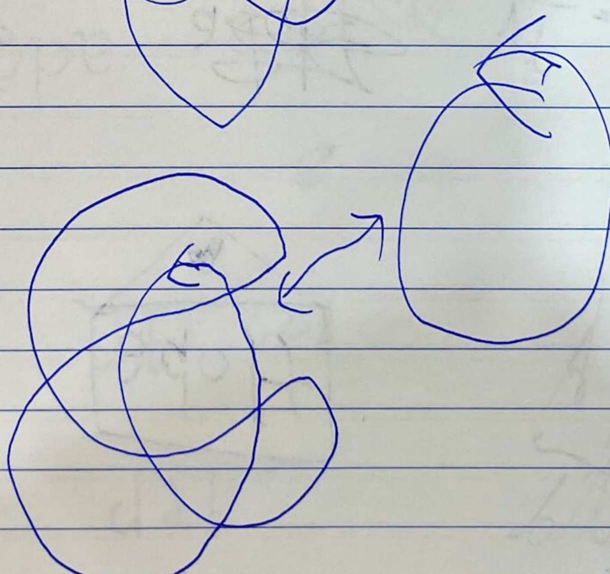
sin sinr sinh

cos cosr cosh









$\frac{4}{3}\pi\int_{-1}^{1}\left(\sqrt{1-x^2}\right)^3 dx$

~~$u = \sin x$~~ $x = \sin u$

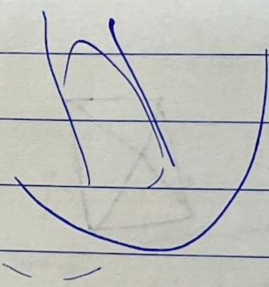
$\frac{d}{d\frac{d}{dd}}$ $\frac{d}{dd}$

$dx = \sqrt{1-x^2}\,du$

$\int (1-x^2)^2\,du$

$\int (\cos u)^4\,du$

$\cos^4$



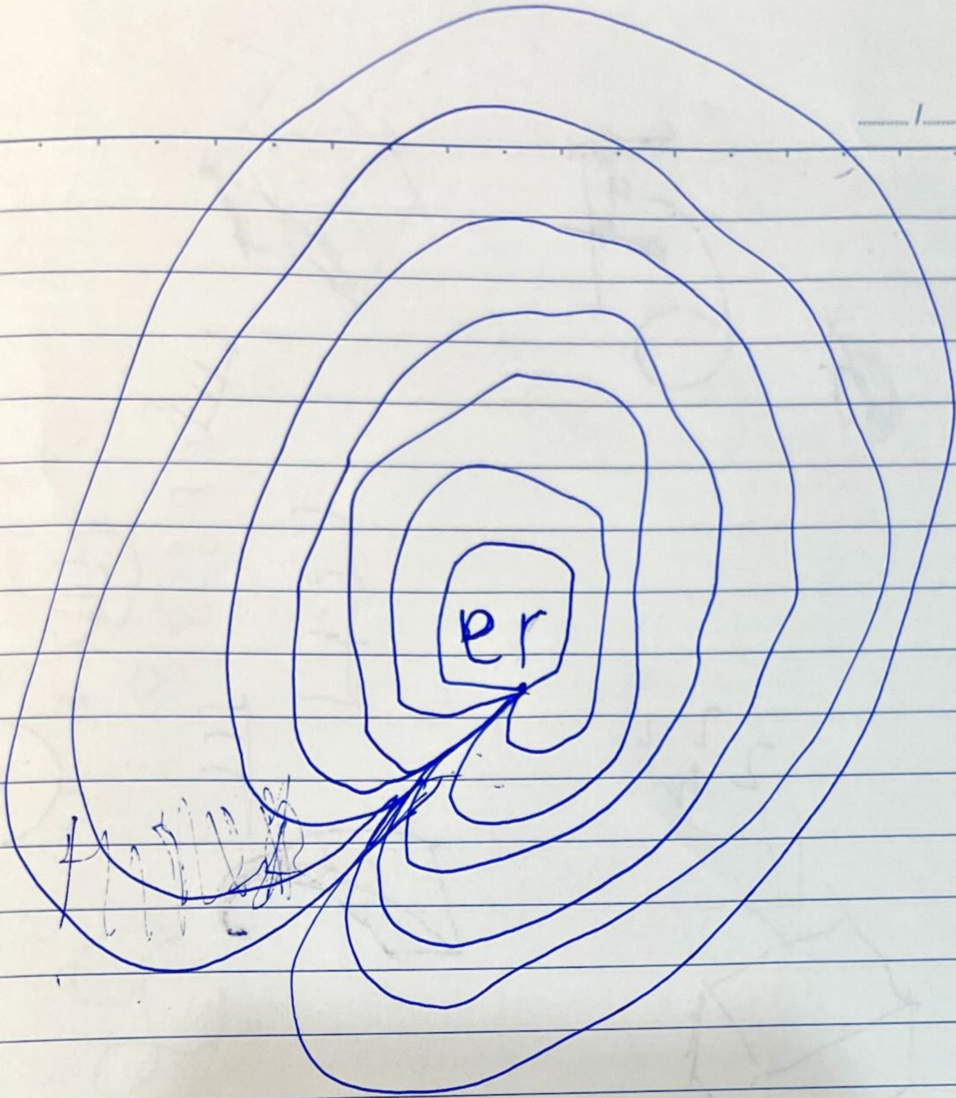
$\frac{d}{d\,\frac{d}{dd}}$ $\frac{d}{dd}$

~~Hope~~

$\frac{d^2\frac{d}{du}}{dd^2}$

$\frac{d^2}{d\,d\frac{d}{dd}} f(x) = f(x)$

$\frac{d}{dx}$



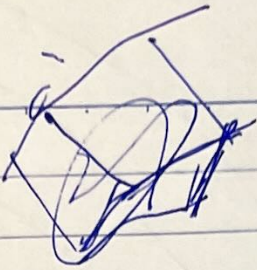
fʃope

dd

ddb

....../....../......

6 9

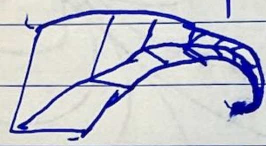


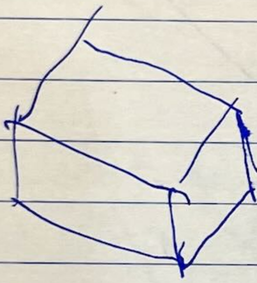
E

E E F

F

2 2 2

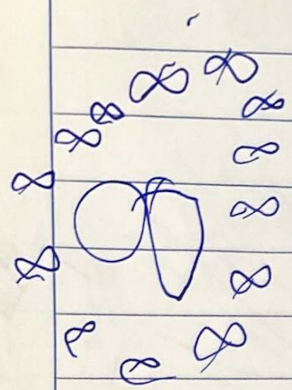


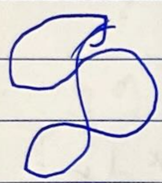


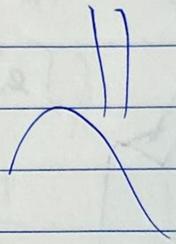
1 2 3 4 5 6 7 8 9 10

0 5 10 15 20

HAI TIEN







$$f_n(x) = 1^n + 2^n + 3^n + \dots + x^n$$

$$= \sum_{i=1}^{x} i^n$$

$$= \cancel{\sum_{i=1}^{x} \left(i(i+1)(i+2)\dots(i+n-1) - i \right.}$$

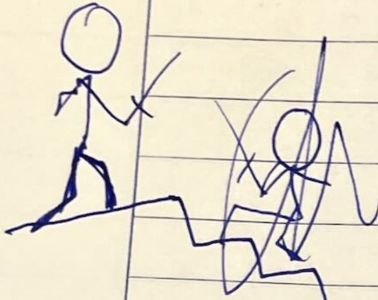
$$= \sum_{i=1}^{x} \left(i(i+1)\dots(i+n-1) - \underbrace{i(i+1)(i+2)\dots(i+n-1) - i^{n}}_{P_n(i)} \right.$$

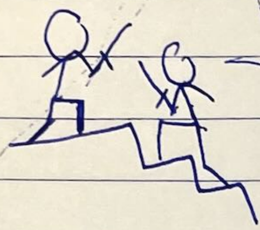
$$= \frac{i(i+1)\dots(i+n)}{n+1} - \sum f_{\dots}(x)$$

$\cancel{\int x^n e^{ax} dx}$ $\int x^n e^{-x} dx = \sum_{i=0}^{\infty} x^i e^{-x} n(n-1)\dots(n-i+1) + C$

$\cancel{\int x^n e^{-x} dx = \sum_{i=0}^{\infty} e^{-x} x^i}$

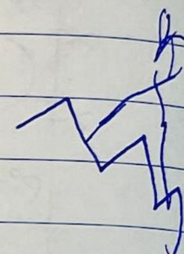
$n \in \mathbb{N}$





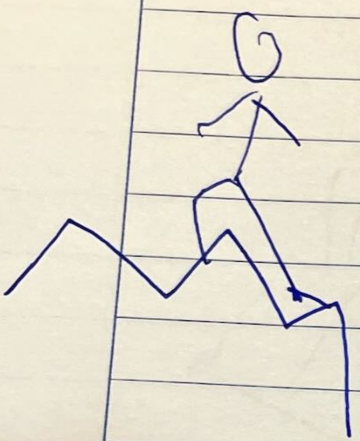
e^x

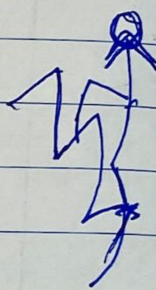
$\cancel{\int x e^x dx}$



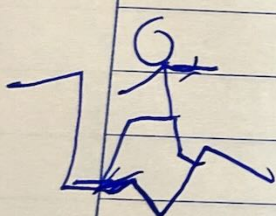
$\int x^n e^x dx$

$\int e^{ax} dx = \frac{e^{ax}}{a} + C$

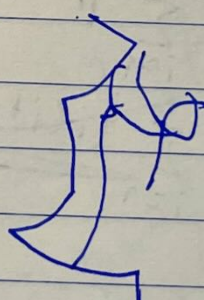




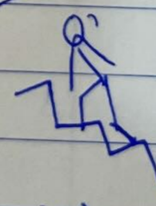
$\frac{\partial^n}{\partial a^n} \int e^{ax} dx = \int x^n e^{ax} dx$



$\frac{d}{da} \frac{e^{ax}}{a}$ $\quad e^{ax} \frac{1}{a}$



$\frac{e^{ax} - a x e^{ax}}{a^2}$ $\quad e^{ax} \frac{1-ax}{a^2}$



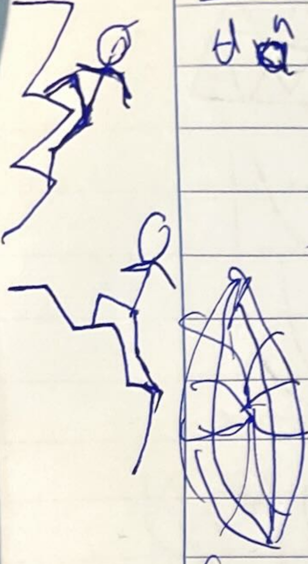
$\int x^n e^{ax} dx = e^{ax} \underbrace{\sum_{i=0}^{\infty} \frac{(-1)^i x^i n(n-1)\dots(n-i+1)}{a^i}}_{\to \infty} + C$

$\int x^n e^{ax} dx = \sum_{i=0}^{\infty} \frac{e^{ax} (-1)^i x^i n(n-1)\dots(n-i+1)}{a^i}$

___/___/______

$$\frac{d^n}{da^n}\frac{e^a}{a} = \cancel{\sum_{i=0}^{\infty} f^{(i)}(a)\,(n+1)n(n-1)\cdots(n-i}$$

$$e^a\sum_{i=0}^{\infty} f_0^{(i)}(a)\,\frac{n(n-1)(n-2)\cdots(n-i+1)}{i!}$$



$$\frac{e^a}{a} \qquad \cancel{e^x + \ldots} \quad e^a\left(\frac{1}{a}-\frac{1}{a^2}\right)$$

$$e^a\left(\frac{1}{a}-\frac{2}{a^2}+\frac{2}{a^3}\right) \qquad f_0(a)=\frac{1}{a}$$

$$e^a\left(\frac{1}{a}-\frac{3}{a^2}+\ldots\right)$$

7 8 9 $\qquad f_0(a)=\frac{1}{a} \qquad \left(\frac{1}{a}\right)^{(i)}$

$$f_{n+1}(a) = f(a)+f'(a) \qquad \frac{(-1)^i\, i!}{a^i}$$

6 9

$$\frac{d^n}{da^n}\frac{e^a}{a} = e^a f_n(a)$$

$$\sum_{i=0}^{\infty}\frac{(-1)^i\, n(n-1)\cdots(n-i+1)}{a^i}$$

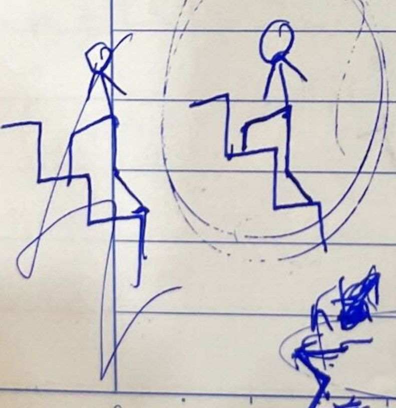
$f_0(a)$

$f_0(a)+f_0'(a)$

$f_0(a)+2f_0'(a)+f_0''(a)$

$f_0(a)+3f_0'(a)+3f_0''(a)+f_0'''(a)$

$$\frac{d^n}{da^n}\frac{e^{ax}}{a} = \sum_{i=0}^{\infty}\frac{(-1)^i x^i\, n(n-1)\cdots(n-i+1)}{a^i}$$



1 0 0 0 0

1 1 0 0 0

1 2 1 0 0 ...

1 3 3 1 0

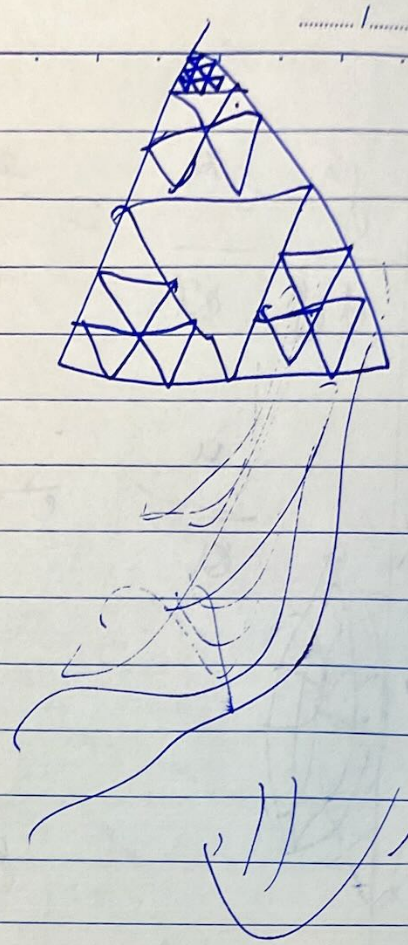
$$\cancel{\sum_{i=0}^{n} f^{(i)}(a)\binom{n}{i}} \qquad \sum_{i=0}^{n} f_0^{(i)}(a)\binom{n}{i}$$

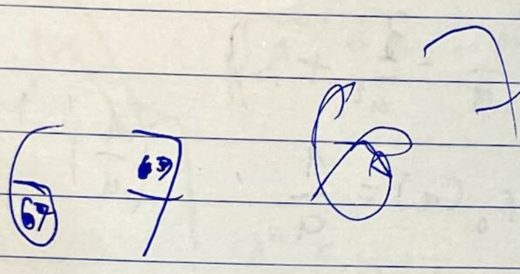
0 5 10 15

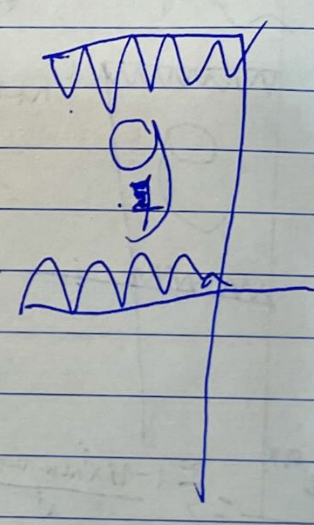
HAI TIEN

$$\frac{e^{x}-e^{-x}}{2}$$

$$\frac{d^n}{dx^n}x^n = n!$$

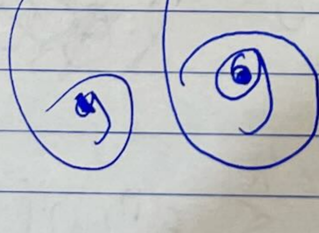


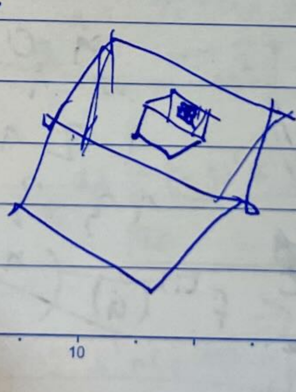




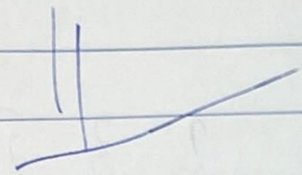
Meow

$$x^n e^{-x}$$



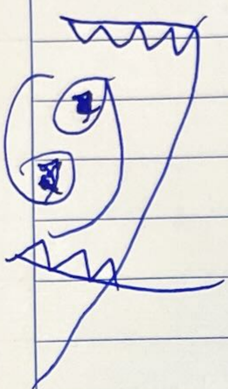


$$\ln n! = \sum_{i=1}^{n} \ln(i)$$



$$\ln(n+1)! - \ln n! = \ln(n+1)$$

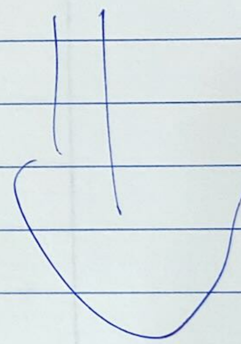
$$\frac{\ln(n+1)! - \ln n!}{\ln n! - \ln(n-1)!} \to 1$$



$$\frac{\ln(n+a)! - \ln n!}{a \ln n} \to 1$$

∞

$$\frac{(n+a)!}{n!}$$



$$(n+1)! = n!(n+1) \approx n\, n!$$

$$(n+a)! \approx n^a n!$$

$$a! = \lim_{n\to\infty} \frac{n^a n!}{(a+1)(a+2)\dots(a+n)}$$

	2	3	5	7
2	X	X	X	X
3	23	X	53	73
5	X	X	X	X
7	X	37	X	X